

Module 1 – Cloud Computing & Enabling Technologies (Complete Q&A; Mindmap)

What is Computing?

- Computing = Managing, Processing, Communicating information.
- Evolution brought: More computational capacity, More efficiency, Less cost, Smaller size.

What are Computing Environments?

- Personal Computing: Single system, local processing, semiconductor-based, reduced size, increased storage.
- Time-Sharing: Multiple users, each gets a time slice, illusion of exclusive use.
- Networked Computing: For resource sharing & communication. Evolved from LANs → ARPANET → Internet. ARPANET used TCP/IP.
- Client-Server: Client requests, Server responds. Server can handle many clients.
- Distributed Computing: Physically separate nodes, network linked, centralized resource manager, components + connectors architecture.
- Grid Computing: Like electricity grid → many nodes, tasks split & distributed, middleware required.

Example of Grid Computing?

- TeraGrid (USA): 11 sites, petaflop capacity, 30PB+ storage.
- 4000+ researchers, 200+ universities.
- CTSS services: single sign-on, job submission, workflow, accounting, compilers, verification.

Advantages of Grid Computing?

- Improved resource utilization, Better performance (parallel processing), Easier collaboration, Increased robustness.

What is Grid Computing?

- Practice of using multiple geographically distributed computers to complete joint tasks.
- Runs on a data grid – computers coordinate to share jobs.

How does Grid Computing Work?

- Specialized software installed on all nodes.
- Manages subtasks, distributes them, gathers outputs.
- Ensures parallel execution & inter-node communication.

Uses of Grid Computing?

- Enables geographically distant experts to collaborate.
- Teams can work on sub-parts of a project.
- Example: Weather prediction → North & South Atlantic regions analyzed separately, results merged.

What is an In-Memory Data Grid (IMDG)?

- Local form of grid computing.
- Nodes store & process data in memory.
- Advantages: very fast access, high throughput, low latency.
- Example: corporate cluster for real-time workloads.

What are the Types of Grids?

1. Computational Grids – high-performance servers connected together.
2. Scavenging Grids – idle desktop CPUs (“cycle scavenging”).
3. Data Grids – manage & transfer large datasets.

What are the Design Issues in Grid Computing?

- Heterogeneity (different OS/architectures).
- Security (trusting node outputs).
- Dynamic nature (nodes joining/leaving).
- Network connectivity (node availability).

Why is Grid Computing compared to an Electric Grid?

- Nodes act as a single entity.
- User doesn't know which node served them.
- Collaborative, often free service.

What is Utility Computing?

- Computing as a utility service like electricity.
- Every connection metered.
- Pay-as-you-use model.

What are Issues/Challenges in Cloud Computing?

- Distance between data & cloud.
- Network latency, bandwidth cost.
- Response time & processing speed.
- Operational cost.
- Security.

What are the Layers of Computing?

- Cloud Layer: Big data, analytics, warehousing.
- Fog Layer: Local micro-data centers, caching.
- Edge Layer: Device-level real-time processing.
- ✓ Edge reduces network traffic.

What is Cloud Computing?

- Combination of Grid + Utility Computing.
- Outsourced to Cloud Service Providers.
- Features: Managed externally, illusion of infinite resources, pay-as-you-go, network accessible.

How did Applications Evolve?

- Standalone Apps: Local, costly, unshareable, frequent updates.
- Web Apps: Client-server, remote, network-dependent, inflexible.
- Cloud Apps: Multi-tenant, elastic, heterogeneous, on-demand, measured use.

Is Cloud Computing always the best choice?

- Not for real-time systems (e.g., driverless cars).
- Better alternative: Edge Computing for instant response.

What are the Features of Edge Computing?

- Advantages: Low latency, fast decisions, privacy, improved security (local encryption).
- Disadvantages: Limited computation power, no global view.
- Example: Museum robots need both local processing & broad context.

What is Fog Computing?

- Between cloud & edge.
- Periodic sync with cloud, caches local info.
- Features: location awareness, low latency, geographical spread, heterogeneity, interoperability, scalability.

What are the Benefits of Fog Computing?

- Low latency.
- Better data control.

- Flexible online/offline storage.
- Bridge between centralized cloud and decentralized local storage.

What are the Benefits of Edge Computing?

- Real-time analysis, immediate action.
- Reduced network traffic.
- Lower operating costs.

What is the Hierarchy of Edge, Fog, and Cloud?

- Edge: Device-level decisions.
- Fog: Local distributed layer.
- Cloud: Central large-scale storage/processing.

What is the Role of IoT with Cloud?

- IoT → huge data → Cloud provides scalable storage.
- Improves security with decentralized storage.
- Cloud platforms (Azure, Google, IBM IoT) provide integration.

What is Mist Computing?

- Lightweight fog computing at network edge.
- Uses microcontrollers/microcomputers → feeds into fog/cloud.

What is Dew Computing?

- Cloud replicated locally → works offline.
- Applicable for personal devices (PC, mobile).
- Example: Dropbox (local copy + sync).

How can businesses scale their systems?

- Horizontal Scaling: Add more systems, no downtime, cheaper long-term, unlimited scale.
- Vertical Scaling: Upgrade one system, downtime possible, costly, limited capacity.

Why is Cloud better for scaling?

- Seasonal demand spikes (holiday traffic).
- Cloud offers auto-scaling, no upfront hardware, pay-as-used.

What is the NIST Definition of Cloud Computing?

- On-demand network access to shared configurable resources (servers, storage, apps, services).
- Resources provisioned/released rapidly with minimal effort.

What are the Essential Features of Cloud Computing?

- On-demand self-service.
- Broad network access.
- Resource pooling (multi-tenancy).
- Rapid elasticity (illusion of infinite resources).
- Measured service (pay-per-use).

What is SLA (Service Level Agreement)?

- Contract between provider & client, covering:
Availability & performance, Data security & privacy, Disaster recovery, Data location & portability, Problem resolution & escalation, Exit strategy.

What are the Enabling Technologies for Cloud?

- Virtualization, Service-Oriented Architecture (SOA), Web 2.0, Browser as a Platform, Parallel Processing.

What is Virtualization?

- Core of cloud computing.
- Creates multiple virtual environments on same hardware.
- Example: Hardware virtualization → multiple OS stacks on one machine, isolated in VMs.

What is Web 2.0?

- Cloud services delivered via the web.
- Interactive, flexible applications using AJAX, XML, web services.
- Examples: Google Docs, Maps, YouTube, Facebook, Twitter, Wikipedia.

What is Service-Oriented Computing?

- Services = main building blocks.
- Features: loosely coupled, reusable, location-transparent, language-independent.
- Benefits: flexible, low-cost, rapid system development.
- Introduces:
 - SaaS (Software as a Service)
 - QoS (Quality of Service) – functional & non-functional attributes of service behavior.